Sayan Karmakar

BS–MS Dual Degree in Chemical Sciences | Minor in Physical Sciences
Indian Institute of Science Education and Research (IISER) Kolkata

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Research Experience

Master's Thesis Student Aug 2026–Present
IISER Kolkata & CSIR-NCL Pune, India *On-site*
Supervisor: Dr. Debangshu Chaudhuri & Dr. Sayan Bagchi (Co-supervisor)

Thesis: *Chiral Supramolecular Assembly and Stereoselective Catalysis Regulated by Water Dynamics in Liquid Condensates*

- Synthesis and Characterization of amino acid functionalized perylene diimide dye molecules.
- Investigating the supra-molecular assembly's structural and spectroscopic properties in molecular condensates using UV-Visible , Circular Dichroism Spectroscopy and Microscopic techniques.
- Studying ultrafast water dynamics inside the condensates using 2DIR spectroscopy on isotopically labelled water.

Visiting Research Intern May–Jul 2026
Ruhr-University Bochum, Germany *On-site*
Supervisor: Prof. Dr. Poul B. Petersen

Project: *Solvation Shell Spectroscopy of Solutes Using UV-Vis Spectrophotometer*

- Carried out UV-Visible spectroscopy experiments to investigate how co-solutes alter the solvation shell of a primary solute.
- Systematically investigated different co-solutes (NaN_3, $NaNO_3$, NaSCN) while maintaining *tert*-butanol as the primary solute.
- Assisted with Sum Frequency Generation (SFG) spectroscopy experiments to study air-water interfaces.

Semester Project Student Aug–Dec 2025
IISER Kolkata, India *On-site*
Supervisor: Dr. Susmita Roy

Course Project: *Calculation of Preferential Interaction Coefficient of Salt Around a Polyelectrolyte*

- Simulated all-atom molecular dynamics of various salt systems (NaCl, KCl, $MgCl_2$) surrounding a 12-bp DNA polyelectrolyte.
- Developed custom Python scripts to process simulation trajectory datasets and compute preferential interaction coefficients.
- Rank-ordered the salt interactions based on computed interaction coefficients to reconstruct the Hofmeister series.

Summer Research Intern May–Jul 2025
CSIR-NCL, Pune, India *On-site*
Supervisor: Dr. Sayan Bagchi

Project: *2D-IR Spectroscopy On Biomolecular Condensates*

- Utilized UV-Visible and FTIR spectroscopy to characterize liquid-liquid phase separation (LLPS) in bovine serum albumin (BSA) protein.
- Operated a ultrafast 2D-IR spectroscopy setup to explore phase-specific dynamics (dense vs. dilute) of BSA condensates.
- Processed and analyzed complex 2D-IR experimental spectral datasets using custom MATLAB pipelines.

Summer Research Intern May–Jul 2024
IISER Kolkata, India *On-site*
Supervisor: Dr. Sangita Sen (Quant-Act Lab)

Project: *Error Mitigation In Quantum Computation*

- Implemented quantum error mitigation protocols including Zero Noise Extrapolation (ZNE) and Read-out Error Mitigation (REM).
- Applied ZNE error mitigation techniques to quantum algorithms estimating the ground state energy of molecular hydrogen (H_2).

Research Interests

Ultrafast Spectroscopy, Supramolecular Chemistry, Biomolecular Condensates, Solvation Dynamics, Molecular Dynamics Simulations.

Education

Indian Institute of Science Education and Research (IISER) Kolkata 2022–2027(Expected)
BS–MS Dual Degree in Chemical Sciences | Minor in Physical Sciences | CGPA: 8.9/10

Nalanda Academy, Kota, Rajasthan Completed 2021
Higher Secondary Education (Class XII) | CBSE | Score: 88.25%

Holy Angels' School, Katwa, West Bengal Completed 2019
Secondary Education (Class X) | ICSE | Score: 96.00%

Honors and Awards

Solvation Scholars Undergraduate Exchange - 2026 Summer Research Programme .

Technical Skills

Programming & Tools: Python, MATLAB, LaTeX, Qiskit, Pennylane

Scientific Software: GROMACS, Linux, Gnuplot, Origin, Microsoft Office Suite

Languages

English: Advanced (CEFR: C1 | TOEFL Score: 5.0/6.0)

Bengali: Native Language

Hindi: Limited Working Proficiency

Extracurricular Activities

Swimming, Playing Football, Video Editing.